C6.2

Continuous Optimization

Lecture notes

(based on the lectures of Coralia Cartis, Oxford)

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Chapter 1

Optimality Conditions for Unconstrained Optimization

1.1 Problem classes and notation

We consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \tag{UP}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is smooth (typically $f \in \mathcal{C}^1$ or $\mathcal{C}^2$).

Definition 1.1 (Local and global minimizers)**.** A point $x^* \in \mathbb{R}^n$ is a *local minimizer* of f if there exists $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in N(x^*, \delta) = \{x : \|x - x^*\| < \delta\}$. It is a *strict* local minimizer if the inequality is strict for $x \neq x^*$, and a *global minimizer* if $f(x^*) \leq f(x)$ for all $x \in \mathbb{R}^n$.

Definition 1.2 (Descent direction)**.** Given $x \in \mathbb{R}^n$ with $\nabla f(x) \neq 0$, a vector $s \in \mathbb{R}^n$ is a *descent direction* for f at x if $\nabla f(x)^\top s < 0$.

Lemma 1.3 (Descent property)**.** *If $f \in \mathcal{C}^1$ and s is a descent direction at x, then there exists $\bar{\alpha} > 0$ such that $f(x + \alpha s) < f(x)$ for all $\alpha \in (0, \bar{\alpha}]$.*

Proof. Since s is a descent direction, $\nabla f(x)^\top s < 0$. By continuity of ∇f, there exists $\bar{\alpha} > 0$ such that

$$\nabla f(x + ts)^\top s < 0$$

for all $t \in [0, \bar{\alpha}]$. Hence, for any $\alpha \in (0, \bar{\alpha}]$, by the mean value theorem there exists $t \in (0, \alpha)$ such that

$$f(x + \alpha s) - f(x) = \alpha \cdot \nabla f(x + ts)^\top s < 0.$$

□

1.2 First-order optimality conditions

Theorem 1.4 (First-order necessary conditions)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$. If x^* is a local minimizer of f, then $\nabla f(x^*) = 0$.*

Proof. Suppose not. Put $s = -\nabla f(x^*)$ and note that s is a descent direction. By Lemma 1.3 there exists $\bar{\alpha} > 0$ such that $f(x^* + \alpha s) < f(x^*)$ for all $\alpha \in (0, \bar{\alpha}]$, contradiction. □

Definition 1.5 (Stationary point)**.** A point x with $\nabla f(x) = 0$ is called a *stationary point* of f.

1.3 Second-order optimality conditions

Theorem 1.6 (Second-order necessary conditions)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$. If x^* is a local minimizer of f, then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$ (positive semidefinite), i.e. $s^\top \nabla^2 f(x^*) s \geq 0$ for all $s \in \mathbb{R}^n$.*

Proof. By Theorem 1.4, $\nabla f(x^*) = 0$. Suppose for contradiction that $s^\top \nabla^2 f(x^*) s < 0$ for some $s \in \mathbb{R}^n$. Since $s \neq 0$ and $f \in \mathcal{C}^2$, there exists $\hat{\alpha} > 0$ such that $s^\top \nabla^2 f(x^* + \alpha s) s < 0$ for all $\alpha \in (0, \hat{\alpha}]$. By Taylor's theorem, for any $\alpha \in (0, \hat{\alpha}]$, there exists $\bar{\alpha} \in (0, \alpha)$ such that

$$f(x^* + \alpha s) - f(x^*) = \frac{\alpha^2}{2} s^\top \nabla^2 f(x^* + \bar{\alpha} s) s < 0,$$

contradiction. □

Theorem 1.7 (Second-order sufficient conditions)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$. If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$ (positive definite), then x^* is a strict local minimizer of f.*

Proof. Since $f \in \mathcal{C}^2$, there exists $\hat{\alpha} > 0$ such that $s^\top \nabla^2 f(x^* + \alpha s) s > 0$ for all $\alpha \in (0, \hat{\alpha}]$ and $s \in \mathbb{R}^n$. By Taylor's theorem, for any $\alpha \in (0, \hat{\alpha}]$, there exists $\bar{\alpha} \in (0, \alpha)$ such that

$$f(x^* + \alpha s) - f(x^*) = \frac{\alpha^2}{2} s^\top \nabla^2 f(x^* + \bar{\alpha} s) s > 0.$$

Thus x^* is a strict local minimizer within the ball $N(x^*, \hat{\alpha})$. □

Example 1.8 (Necessary $\not\Rightarrow$ sufficient)**.** The conditions are different:

- $f(x) = x^3$, $x^* = 0$: satisfies the second-order necessary condition ($f''(0) = 0 \geq 0$) but is not a local minimizer.
- $f(x) = x^4$, $x^* = 0$: is a strict local minimizer but the second-order sufficient condition *fails* ($f''(0) = 0 \not\succ 0$).

The gap between necessary and sufficient is the case $\nabla^2 f(x^*) \succeq 0$ but not $\succ 0$, in which case optimality cannot be decided from ∇f and $\nabla^2 f$ alone.

1.4 Convex problems

For convex problems, the picture simplifies dramatically: *every local minimizer is global*, and the first-order necessary condition $\nabla f(x^*) = 0$ becomes *sufficient*. This is one of the main reasons convex optimization is algorithmically tractable.

Definition 1.9 (Convex function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is *convex* if its domain is convex and, for all x, y in its domain and all $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y)$.

Theorem 1.10 (Optimality for convex problems)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and $\mathcal{C}^1$. Then x^* is a global minimizer of f if and only if $\nabla f(x^*) = 0$.*

Proof. If x^* is a global minimizer, then by Theorem 1.4, $\nabla f(x^*) = 0$. It remains to show the converse.

We first show that if x^* is a local minimizer, then it is a global minimizer. Suppose not. There exists $y \in \mathbb{R}^n$ such that $f(y) < f(x^*)$. Put $s = y - x^*$. By convexity, for $\lambda \in (0, 1)$,

$$f(x^* + \lambda s) = f((1 - \lambda) x^* + \lambda y) \leq (1 - \lambda) f(x^*) + \lambda f(y) < f(x^*).$$

But then x^* is a local minimizer, so there exists $\delta > 0$ such that $f(x^* + \lambda s) \leq f(x^*)$ for all $\lambda \in (0, \delta)$, contradiction.

[Proof to be filled in.] □

Chapter 2

Linesearch Methods

A linesearch method generates iterates $x^{k+1} = x^k + \alpha_k s^k$ where s^k is a descent direction and $\alpha_k > 0$ is a stepsize. The philosophy is "liberal in the choice of search direction, keeping bad behaviour in control by choice of α_k". Choosing α_k properly is essential: too long and the method may overshoot, too short and progress is negligible.

2.1 The generic linesearch method

Algorithm 2.1 (Generic Linesearch Method (GLM))**.** Choose $\varepsilon > 0$ and $x^0 \in \mathbb{R}^n$. For $k \geq 0$:

1. If $\left\|\nabla f(x^k)\right\| \leq \varepsilon$, stop.
2. Compute a descent direction s^k with $\nabla f(x^k)^\top s^k < 0$.
3. Compute a stepsize $\alpha_k > 0$ along s^k such that $f(x^k + \alpha_k s^k) < f(x^k)$.
4. Set $x^{k+1} := x^k + \alpha_k s^k$.

2.2 Stepsize rules: the Armijo condition

Choices of α_k. The conceptually simplest choice is the *exact linesearch* $\alpha_k := \arg\min_{\alpha>0} f(x^k + \alpha s^k)$, but this is computationally expensive for nonlinear f. Inexact linesearches replace it with a cheap rule that guarantees the stepsize is "not too long" and "not too short".

The *Armijo condition* demands sufficient decrease: α is acceptable provided the actual decrease is at least a fraction β of what the linear model predicts. In practice β is taken small, e.g. $\beta = 0.1$ or even $\beta = 0.001$, to keep the rule mild.

For $\beta \in (0, 1)$ the *Armijo (sufficient decrease) condition* at iteration k is

$$f(x^k + \alpha s^k) \leq f(x^k) + \beta\alpha\, \nabla f(x^k)^\top s^k. \tag{ac}$$

Remark 2.2 (Wolfe conditions)**.** The Armijo condition prevents α_k from being too long, but on its own admits arbitrarily small α_k. The *Wolfe conditions* add a second ("curvature") inequality $\nabla f(x^k + \alpha s^k)^\top s^k \geq \eta\, \nabla f(x^k)^\top s^k$ with $\eta \in (\beta, 1)$, ruling out steps that are too short. Wolfe linesearches are needed for some methods (e.g. BFGS, see Chapter 5) to ensure the curvature condition $(\delta^k)^\top \gamma^k > 0$.

Algorithm 2.3 (Backtracking-Armijo (bArmijo) linesearch)**.** Choose $\alpha^{(0)} > 0$, $\tau \in (0, 1)$, $\beta \in (0, 1)$. Set $i := 0$. While (ac) fails with $\alpha = \alpha^{(i)}$, set $\alpha^{(i+1)} := \tau\alpha^{(i)}$ and $i := i + 1$. Set $\alpha_k := \alpha^{(i)}$.

Lemma 2.4 (Range of admissible stepsizes)**.** *Let $f \in \mathcal{C}^1$ with ∇f Lipschitz continuous with constant L. At iteration k of GLM, the Armijo condition* (ac) *is satisfied for all $\alpha \in [0, \alpha_{\max}^k]$, where*

$$\alpha_{\max}^k = \frac{(\beta - 1)\, \nabla f(x^k)^\top s^k}{L \left\| s^k \right\|^2}.$$

Proof. First note that $\alpha_{\max}^k > 0$ as $\beta < 1$ and s^k is a descent direction. Let s^k be a descent direction and let $\alpha \in [0, \alpha_{\max}^k]$. By Taylor's theorem, there exists $\tilde{\alpha} \in (0, \alpha)$ such that

$$\begin{aligned}
f(x^k + \alpha s^k) &= f(x^k) + \alpha \nabla f(x^k + \tilde{\alpha} s^k)^\top s^k \\
&= f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha \left[\nabla f(x^k + \tilde{\alpha} s^k) - \nabla f(x^k)\right]^\top s^k \\
&\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha L \left\| x^k - \tilde{\alpha} s^k - x^k \right\| \cdot \left\| s^k \right\| \\
&\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha^2 L \left\| s^k \right\|^2 \\
&\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha^2 (\beta - 1) \nabla f(x^k)^\top s^k \\
&\leq f(x^k) + \beta \alpha \nabla f(x^k)^\top s^k,
\end{aligned}$$

which satisfies the Armijo condition. □

Lemma 2.5 (Lower bound on bArmijo stepsize)**.** *Under the conditions of Lemma 2.4, the bArmijo stepsize α_k satisfies $\alpha_k \geq \min\{\alpha^{(0)}, \tau \alpha_{\max}^k\}$ for all $k \geq 0$.*

Proof. Note that $a_k = a^{(0)}$ if bArmijo terminates at the start. Suppose this is not the case. By Lemma 2.4, bArmijo terminates whenever $\alpha \leq \alpha_{\max}^k$. Let $i - 1$ be the last iteration before termination, so $\alpha^{(i-1)} > \alpha_{\max}^k$ and $\tau \alpha^{(i-1)} = \alpha^{(i)} \leq \alpha_{\max}^k$. But then

$$\alpha_k = \alpha^{(i)} = \tau \alpha^{(i-1)} > \tau \alpha_{\max}^k.$$

□

2.3 Global convergence of GLM

Theorem 2.6 (Global convergence of GLM with bArmijo)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below by f_{low}, and ∇f be Lipschitz continuous. Apply GLM with bArmijo linesearch ($\varepsilon := 0$). Then either there exists $\ell \geq 0$ with $\nabla f(x^\ell) = 0$, or*

$$\lim_{k \to \infty} \min \left\{ \frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}, |\nabla f(x^k)^\top s^k| \right\} = 0.$$

Proof. Suppose $\nabla f(x^k) \neq 0$ for all $k \geq 0$. Then Armijo condition with $x^{k+1} = x^k + \alpha_k s^k$ gives

$$f(x^{k+1}) \leq f(x^k) + \beta \alpha_k \nabla f(x^k)^\top s^k \implies f(x^k) - f(x^{k+1}) \geq \beta \alpha_k |\nabla f(x^k)^\top s^k|.$$

for all $k \geq 0$. By the telescoping sum,

$$f(x^0) - f(x^{k+1}) \geq \sum_{i=0}^{k} \beta \alpha_i |\nabla f(x^i)^\top s^i|.$$

Since f is bounded below by f_{low},

$$\infty > f(x^0) - f_{\text{low}} \geq \sum_{i=0}^{\infty} \beta \alpha_i |\nabla f(x^i)^\top s^i|.$$

Thus, $\alpha_k|\nabla f(x^k)^\top s^k| \to 0$ as $k \to \infty$. Recall from Lemma 2.5 that $\alpha_k \geq \min\{\alpha^{(0)}, \tau\alpha^k_{\max}\}$ for all $k \geq 0$. Define

$$\mathcal{K}_1 \coloneqq \{k : \alpha^{(0)} \geq \tau\alpha^k_{\max}\}, \qquad \mathcal{K}_2 \coloneqq \{k : \alpha^{(0)} < \tau\alpha^k_{\max}\}.$$

By Lemma 2.5, for $k \in \mathcal{K}_1$,

$$\alpha_k|\nabla f(x^k)^\top s^k| \geq \frac{(1-\beta)\tau}{L} \cdot \left(\frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}\right)^2 \geq 0.$$

But then $\alpha_k|\nabla f(x^k)^\top s^k| \to 0$ as $k \to \infty$ in $\mathcal{K}_1$, so

$$\lim_{\substack{k\to\infty \\ k\in\mathcal{K}_1}} \frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|} \to 0.$$

On the other hand, by Lemma 2.5, for $k \in \mathcal{K}_2$ we have $a_k \geq \alpha^{(0)}$, so

$$0 \leq \lim_{\substack{k\to\infty \\ k\in\mathcal{K}_2}} |\nabla f(x^k)^\top s^k| \leq \frac{1}{\alpha^{(0)}} \cdot \lim_{\substack{k\to\infty \\ k\in\mathcal{K}_2}} \alpha_k|\nabla f(x^k)^\top s^k| \to 0.$$

Combining both cases, we have

$$\lim_{k\to\infty} \min\left\{\frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}, \nabla f(x^k)^\top s^k|\right\} = 0.$$

□

Remark 2.7. Writing $\cos\theta_k = \frac{-\nabla f(x^k)^\top s^k}{\|\nabla f(x^k)\|\|s^k\|}$, Theorem 2.6 gives $\left\|\nabla f(x^k)\right\| \cos\theta_k \min\{1, \left\|s^k\right\|\} \to 0$. Hence to ensure $\left\|\nabla f(x^k)\right\| \to 0$ it is not enough that s^k be descent: the angle condition $\cos\theta_k \geq \delta > 0$ for some δ is required.

Chapter 3

The Steepest Descent Method

The steepest descent (SD) method is the simplest concrete instance of GLM, obtained by choosing $s^k = -\nabla f(x^k)$. It is a *first-order method*: only gradient information is used.

3.1 The steepest descent direction

Definition 3.1 (Steepest descent direction)**.** The *steepest descent (SD) direction* at x^k is $s^k_{\mathrm{SD}} := -\nabla f(x^k)$.

Lemma 3.2 (Optimality of $-\nabla f$)**.** *The steepest descent direction is the unique global solution of*

$$\min_{s\in\mathbb{R}^n} \; f(x^k) + s^\top \nabla f(x^k) \quad \textit{subject to} \quad \|s\| = \left\|\nabla f(x^k)\right\|.$$

Proof. Note that

$$\arg\min_{s\in\mathbb{R}^n} f(x^k) + s^\top \nabla f(x^k) = \arg\min_{s\in\mathbb{R}^n} s^\top \nabla f(x^k).$$

By Cauchy-Schwarz inequality,

$$|s^\top \nabla f(x^k)| \le \|s\| \left\|\nabla f(x^k)\right\| = \left\|\nabla f(x^k)\right\|^2,$$

with equality if and only if $s = \pm\nabla f(x^k)$. In particular,

$$s^\top \nabla f(x^k) \ge -\left\|\nabla f(x^k)\right\|^2,$$

which is only achieved when $s = -\nabla f(x^k)$. □

3.2 Global convergence of steepest descent

Theorem 3.3 (Global convergence of SD with bArmijo)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below with ∇f Lipschitz continuous. Apply GLM with $s^k = -\nabla f(x^k)$ and bArmijo linesearch. Then either there exists ℓ with $\nabla f(x^\ell) = 0$, or $\left\|\nabla f(x^k)\right\| \to 0$ as $k \to \infty$.*

Proof. Suppose there are no ℓ with $\nabla f(x^\ell) = 0$. Then by Theorem 2.6,

$$\lim_{k\to\infty} \min\left\{ \frac{\left\|\nabla f(x^k)\right\|^2}{\|-\nabla f(x^k)\|}, \left\|\nabla f(x^k)\right\| \right\} = \left\|\nabla f(x^k)\right\| = 0.$$

□

3.3 Local rate of convergence

Definition 3.4 (Rates of convergence)**.** A sequence $\{x^k\} \subset \mathbb{R}^n$ converges to x^* with rate $p \geq 1$ if there exist $\rho > 0$ and k_0 with $\left\|x^{k+1} - x^*\right\| \leq \rho \left\|x^k - x^*\right\|^p$ for all $k \geq k_0$. The cases $p = 1$ ($\rho < 1$) and $p = 2$ are called *linear* and *quadratic* convergence respectively. The convergence is *superlinear* if $\left\|x^{k+1} - x^*\right\| / \left\|x^k - x^*\right\| \to 0$.

Theorem 3.5 (Local linear convergence of SD)**.** *Let $f \in \mathcal{C}^2$ and let x^* be a local minimizer of f with $\nabla^2 f(x^*) \succ 0$ and eigenvalues $0 < \lambda^*_{\min} \leq \lambda^*_{\max}$. Apply SD with exact linesearch. If $x^k \to x^*$, then the convergence is linear with factor*

$$\rho \leq \frac{\kappa(x^*) - 1}{\kappa(x^*) + 1}, \qquad \kappa(x^*) = \frac{\lambda^*_{\max}}{\lambda^*_{\min}}.$$

□

3.4 Summary of steepest descent

- **Inexpensive:** only first derivatives required.
- **Strong global convergence** under weak assumptions, with second-order optimality guarantees for the generated solution (see e.g. saddle-point avoidance properties in Lecture 5).
- **Scale-dependent:** a poor scaling of variables produces a steepest-descent direction that gives little progress. Newton's method does not have this defect (Theorem 4.4).
- **Slow asymptotic convergence**: when $\kappa(x^*)$ is large, $\rho_{SD} = (\kappa - 1)/(\kappa + 1)$ is very close to 1 and the iterates "zig-zag". SD is therefore not used in general-purpose software, but it remains useful for certain very well-conditioned problems with structure (e.g. compressed sensing).

Chapter 4

Newton's Method

Newton's method for unconstrained minimization can be derived in two equivalent ways:

1. *Quadratic model.* The Newton step s^k minimizes the second-order Taylor approximation of f at x^k (when the Hessian is positive definite). Compare this with steepest descent, whose direction minimizes the first-order Taylor approximation under a norm constraint.
2. *Newton's root-finding.* Stationarity is the equation $\nabla f(x) = 0$, an $n \times n$ nonlinear system. Applying Newton's root-finding scheme to $r(x) := \nabla f(x)$ produces $x^{k+1} = x^k - [J(x^k)]^{-1} r(x^k) = x^k - [\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$, recovering the Newton iterate.

4.1 The Newton direction

Definition 4.1 (Newton direction)**.** For $f \in \mathcal{C}^2$ with $\nabla^2 f(x^k)$ nonsingular, the *Newton direction* at x^k is the solution s^k of

$$\nabla^2 f(x^k)\, s^k = -\nabla f(x^k). \tag{4.1}$$

The *(pure) Newton iteration* sets $x^{k+1} := x^k + s^k$.

Lemma 4.2 (Descent property of the Newton direction)**.** *If $\nabla^2 f(x^k) \succ 0$ and $\nabla f(x^k) \neq 0$, then the Newton direction is a descent direction. Moreover, s^k is the unique minimizer of the second-order Taylor model $q_k(s) := f(x^k) + s^\top \nabla f(x^k) + \frac{1}{2} s^\top \nabla^2 f(x^k) s$.*

Proof. Put $H_k := \nabla^2 f(x^k)$ and $g_k := \nabla f(x^k)$. Since $H_k \succ 0$, it is nonsingular and the Newton equation gives

$$s^k = -H_k^{-1} g_k.$$

Also $H_k^{-1} \succ 0$, so, as $g_k \neq 0$,

$$g_k^\top s^k = -g_k^\top H_k^{-1} g_k < 0,$$

which makes s^k a descent direction.

It remains to prove the model-minimizing property. Completing the square, we may write

$$q_k(s) = f(x^k) - \frac{1}{2} g_k^\top H_k^{-1} g_k + \frac{1}{2}(s + H_k^{-1} g_k)^\top H_k (s + H_k^{-1} g_k).$$

The last term is nonnegative, and it vanishes if and only if $s = -H_k^{-1} g_k = s^k$. Hence s^k is the unique global minimizer of q_k. □

4.2 Local convergence of pure Newton

Theorem 4.3 (Local quadratic convergence of pure Newton)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$, $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ nonsingular, and $\nabla^2 f$ locally Lipschitz continuous at x^*. If x^{k_0} is sufficiently close to x^* for some $k_0 \geq 0$, then x^k is well-defined for all $k \geq k_0$ and $x^k \to x^*$ at a quadratic rate.*

Proof. Write $g(x) := \nabla f(x)$, $H(x) := \nabla^2 f(x)$, and $e^k := x^k - x^*$. Since $H(x^*)$ is nonsingular and H is continuous, there is a radius $\rho > 0$ such that $H(x)$ is nonsingular for $x \in N(x^*, \rho)$. Shrinking ρ if necessary, local Lipschitz continuity gives a constant $L > 0$ such that

$$\|H(y) - H(x)\| \le L \|y - x\| \quad \text{for all } x, y \in N(x^*, \rho),$$

and continuity of $x \mapsto H(x)^{-1}$ gives a constant $M > 0$ such that

$$\left\|H(x)^{-1}\right\| \le M \quad \text{for all } x \in N(x^*, \rho).$$

For any $x \in N(x^*, \rho)$, Taylor's theorem for g with Lipschitz derivative gives

$$g(x^*) = g(x) + H(x)(x^* - x) + r(x), \qquad \|r(x)\| \le \frac{L}{2} \|x - x^*\|^2 .$$

Since $g(x^*) = 0$, the pure Newton iterate $N(x) := x - H(x)^{-1} g(x)$ satisfies

$$N(x) - x^* = H(x)^{-1}\big(H(x)(x - x^*) - g(x)\big) = H(x)^{-1} r(x).$$

Hence

$$\|N(x) - x^*\| \le \frac{ML}{2} \|x - x^*\|^2 .$$

Choose $\delta \in (0, \rho)$ so small that $\frac{ML}{2}\delta \le \frac{1}{2}$. If $x^{k_0} \in N(x^*, \delta)$, the above estimate shows inductively that every Newton iterate is well-defined and remains in $N(x^*, \delta)$, since

$$\left\|x^{k+1} - x^*\right\| \le \frac{ML}{2} \left\|x^k - x^*\right\|^2 \le \frac{1}{2} \left\|x^k - x^*\right\| .$$

Therefore $x^k \to x^*$, and the sharper estimate

$$\left\|x^{k+1} - x^*\right\| \le \frac{ML}{2} \left\|x^k - x^*\right\|^2$$

gives quadratic convergence. □

Theorem 4.4 (Scale invariance of Newton)**.** *Newton's method (with or without linesearch) is scale invariant under linear transformations of the variables. Explicitly, if $A \in \mathbb{R}^{n\times n}$ is nonsingular, $y = Ax$, and $\bar{f}(y) := f(A^{-1}y)$, then the Newton direction s_y for $\bar{f}$ at y satisfies $y + \alpha s_y = A(x + \alpha s_x)$.*

Proof. Note that

$$\nabla \bar{f}(y) = (A^{-1})^\top \nabla f(x), \quad \nabla^2 \bar{f}(y) = (A^{-1})^\top \nabla^2 f(x) A^{-1}.$$

Thus,

$$s_y = -\big[\nabla^2 \bar{f}(y)\big]^{-1} \nabla \bar{f}(y) = \big(A\big[\nabla^2 f(x)\big]^{-1} A^\top\big)\big((A^{-1})^\top \nabla f(x)\big) = A s_x.$$

□

Remark 4.5 (Disadvantages of Newton's method)**.** Despite its fast local convergence, the pure Newton iteration suffers several serious drawbacks:

- s^k is not well-defined when $\nabla^2 f(x^k)$ is singular.
- s^k may not be a descent direction when $\nabla^2 f(x^k)$ is not positive definite.
- Pure Newton may be attracted to local maxima or saddle points (the local convergence Theorem 4.3 only requires $\nabla^2 f(x^*)$ nonsingular, not positive definite).
- Pure Newton may fail to converge if x^0 is far from a solution; for example, applied to $f(x) = -x^6/6 + x^4/4 + 2x^2$ from $x^0 = 1$, it cycles between ± 1 (neither stationary).

This is why a linesearch (or trust-region) safeguard is needed to make Newton's method globally convergent.

4.3 Linesearch Newton (damped Newton) method

Algorithm 4.6 (Linesearch Newton method)**.** Choose $\varepsilon > 0$ and $x^0 \in \mathbb{R}^n$. While $\left\|\nabla f(x^k)\right\| > \varepsilon$: solve $\nabla^2 f(x^k)s^k = -\nabla f(x^k)$, then set $x^{k+1} = x^k + \alpha_k s^k$ with $\alpha_k \in (0, 1]$ chosen by, e.g., bArmijo linesearch.

Theorem 4.7 (Local convergence of linesearch Newton)**.** *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$ with $\nabla^2 f$ Lipschitz continuous and positive definite at the iterates. Apply Newton's method with bArmijo linesearch using $\beta < 1/2$ and $\alpha^{(0)} = 1$. If $x^k \to x^*$ with $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$, then $\alpha_k = 1$ for all k sufficiently large and the convergence is asymptotically quadratic.*

Proof. Put $g_k := \nabla f(x^k)$, $H_k := \nabla^2 f(x^k)$, and let $s^k = -H_k^{-1} g_k$ be the Newton direction. Since $x^k \to x^*$ and $H(x^*) \succ 0$, there exist $m > 0$ and k_0 such that

$$H_k \succeq mI \quad \text{for all } k \geq k_0.$$

In particular, for $k \geq k_0$,

$$g_k^\top s^k = -(s^k)^\top H_k s^k \leq -m \left\|s^k\right\|^2.$$

Also $g_k \to 0$ and H_k^{-1} is bounded for large k, so $s^k \to 0$.

Let L be a Lipschitz constant for $\nabla^2 f$ on a neighbourhood of x^* containing x^k and $x^k + s^k$ for all sufficiently large k. Taylor's theorem with Lipschitz Hessian gives

$$f(x^k + s^k) \leq f(x^k) + g_k^\top s^k + \frac{1}{2}(s^k)^\top H_k s^k + \frac{L}{6}\left\|s^k\right\|^3.$$

Using $H_k s^k = -g_k$, this becomes

$$f(x^k + s^k) \leq f(x^k) + \frac{1}{2} g_k^\top s^k + \frac{L}{6}\left\|s^k\right\|^3.$$

Since $\beta < 1/2$ and $s^k \to 0$, for all sufficiently large k,

$$\frac{L}{6}\left\|s^k\right\| \leq \left(\frac{1}{2} - \beta\right) m.$$

Therefore

$$\frac{L}{6}\left\|s^k\right\|^3 \leq \left(\frac{1}{2} - \beta\right)(s^k)^\top H_k s^k = \left(\frac{1}{2} - \beta\right)(-g_k^\top s^k),$$

and hence

$$f(x^k + s^k) \leq f(x^k) + \beta\, g_k^\top s^k.$$

Thus the full step $\alpha = 1$ satisfies the Armijo condition. Since bArmijo starts with $\alpha^{(0)} = 1$, it accepts $\alpha_k = 1$ for all sufficiently large k.

From that point on the method is pure Newton. The hypotheses of Theorem 4.3 hold at x^*, so the tail of the sequence satisfies

$$\left\|x^{k+1} - x^*\right\| \leq C\left\|x^k - x^*\right\|^2$$

for some constant $C > 0$ and all sufficiently large k. This is asymptotic quadratic convergence. $\square$

4.4 Global convergence with general second-order models

In GLM, let B_k be symmetric positive definite for all k, and let s^k solve

$$B_k s^k = -\nabla f(x^k). \tag{4.2}$$

Theorem 4.8 (Global convergence with bounded B_k)**.** *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below with ∇f Lipschitz continuous. Apply GLM with s^k from (4.2) and bArmijo linesearch ($\varepsilon := 0$). Suppose the eigenvalues of B_k are uniformly bounded above and uniformly bounded away from zero, independently of k. Then either there exists ℓ with $\nabla f(x^\ell) = 0$, or $\left\|\nabla f(x^k)\right\| \to 0$ as $k \to \infty$.*

Proof. Write $H_k = \nabla^2 f(x^k)$ and $g_k = \nabla f(x^k)$. Note that B_k is nonsingular, so we may write $s^k = -B_k^{-1} g_k$. Assume that $g_k \neq 0$ for all $k \geq 0$. Then by Theorem 2.6

$$\lim_{k\to\infty} \min\left\{ \frac{|g_k^\top s^k|}{\|s^k\|}, |g_k^\top s^k| \right\} = 0. \tag{4.3}$$

We are left to show that this implies that $\left\|\nabla f(x^k)\right\| \to 0$ as $k \to \infty$.

Since B_k is symmetric, B_k admits real eigenvalues by the spectral theorem. Since the eigenvalues are uniformly bounded above and away from zero, there exists constants $\lambda_{\max}$ and $\lambda_{\min}$ such that $\lambda_{\max} \geq \lambda_{\max}(B_k) \geq \lambda_{\min}(B_k) \geq \lambda_{\min}$ for all $k \geq 0$. In particular, for any $s \in \mathbb{R}^n$ and any B_k,

$$\lambda_{\min} \|s\|^2 \leq \lambda_{\min}(B_k) \|s\|^2 \leq s^\top B_k s \leq \lambda_{\max}(B_k) \|s\|^2 \leq \lambda_{\max} \|s\|^2 \implies \lambda_{\min} \leq \frac{s^\top B_k s}{\|s\|^2} \leq \lambda_{\max}.$$

Since ∇f is Lipschitz continuous, $\|H_k\|$ is uniformly bounded above. Thus, by possibly enlarging $\lambda_{\max}$, we have $\lambda_{\max}(H_k) \leq \lambda_{\max}$ for all $k \geq 0$.

By definition of the s^k and $g_k \neq 0$,

$$|g_k^\top s^k| = |g_k^\top B_k^{-1} g_k| \geq \lambda_{\min}(B_k^{-1}) \|g_k\|^2 \geq \frac{\|g_k\|^2}{\lambda_{\max}}.$$

On the other hand,

$$\left\|s^k\right\|^2 = g_k^\top B_k^{-2} g_k \leq \lambda_{\max}(B^{-2}) \|g_k\|^2 \leq \frac{\|g_k\|^2}{\lambda_{\min}^2} \implies \frac{1}{\|s^k\|} \geq \frac{\lambda_{\min}}{\|g_k\|}.$$

Plugging into (4.3), we have

$$\lim_{k\to\infty} \min\left\{ \frac{\lambda_{\min}}{\lambda_{\max}} \|g_k\|, \frac{\|g_k\|^2}{\lambda_{\max}} \right\} \leq \lim_{k\to\infty} \min\left\{ \frac{|g_k^\top s^k|}{\|s^k\|}, |g_k^\top s^k| \right\} \to 0 \implies \lim_{k\to\infty} g_k = 0.$$

□

Chapter 5

Quasi-Newton Methods and Nonlinear Least-Squares

Newton's method has fast local convergence but each iteration is expensive: forming $\nabla^2 f(x^k)$ costs $O(n^2)$ second-derivative evaluations, and solving the linear system costs $O(n^3)$. *Quasi-Newton methods* aim for a sweet spot: faster than steepest descent, but cheaper per iteration than Newton, by using *secant approximations* $B_k \approx \nabla^2 f(x^k)$ built from gradient information already computed.

The design principles for B_{k+1} form a "wish list":

- built from already-computed quantities $\nabla f(x^{k+1}), \nabla f(x^k), \ldots, B_k, s^k$;
- symmetric, nonsingular, and ideally positive definite;
- "close" to B_k and a "cheap" update of B_k;
- asymptotically, $B_k \to \nabla^2 f(x^*)$.

No single update satisfies the entire list; different methods make different trade-offs.

Remark 5.1 (Approximating the Hessian by finite differences)**.** An alternative when second derivatives are unavailable is to approximate $[\nabla^2 f(x)]e_i \approx \frac{1}{h}[\nabla f(x+he_i) - \nabla f(x)]$ for $i = 1, \ldots, n$, costing $n+1$ gradient evaluations. Choice of h is delicate (too large $\to$ inaccurate; too small $\to$ cancellation errors).

5.1 The secant equation

Interpretation. The secant equation requires the new model to predict correctly the change in gradient: in the quadratic case $f(x) = g^\top x + \frac{1}{2}x^\top Hx$, $\nabla f(x^{k+1}) - \nabla f(x^k) = H(x^{k+1} - x^k)$, so $B_{k+1} = H$ satisfies the equation. More generally, the change in gradient encodes information about the Hessian (cf. finite differences).

In quasi-Newton methods, $\nabla^2 f(x^{k+1})$ is approximated by a symmetric matrix B_{k+1} satisfying the *secant equation*

$$B_{k+1}\,\delta^k = \gamma^k, \qquad \delta^k := x^{k+1} - x^k, \;\; \gamma^k := \nabla f(x^{k+1}) - \nabla f(x^k). \tag{5.1}$$

Lemma 5.2 (Quadratic case)**.** *If $f(x) = g^\top x + \frac{1}{2}x^\top Hx$, then* (5.1) *is satisfied with $B_{k+1} = \nabla^2 f = H$.*

Proof. [Proof to be filled in.] □

5.2 The BFGS update

There are infinitely many ways to satisfy the secant equation. One choice is the *symmetric rank-1 (SR1)* update $B_{k+1} = B_k + u^k(u^k)^\top$, which is cheap ($O(n^2)$ per iteration via the Sherman–Morrison–Woodbury formula) but may fail to be positive definite, leading to non-descent directions. The *BFGS* update (Broyden, Fletcher, Goldfarb, Shanno, independently 1970) is a rank-2 update that preserves positive definiteness.

Definition 5.3 (BFGS update)**.** The *BFGS* update is the rank-2 symmetric update

$$B_{k+1} = B_k - \frac{B_k \delta^k (\delta^k)^\top B_k}{(\delta^k)^\top B_k \delta^k} + \frac{\gamma^k (\gamma^k)^\top}{(\delta^k)^\top \gamma^k}. \tag{5.2}$$

Theorem 5.4 (Properties of the BFGS update)**.** *If B_k is symmetric positive definite and $(\delta^k)^\top \gamma^k > 0$, then the BFGS update* (5.2) *is well-defined, symmetric and positive definite, and satisfies the secant equation* (5.1)*.*

Proof. [Proof to be filled in.] □

Remark 5.5. The condition $(\delta^k)^\top \gamma^k > 0$ is guaranteed by use of the Wolfe linesearch.

Theorem 5.6 (Convergence of BFGS)**.** *Under suitable conditions on f and using the Wolfe linesearch, the BFGS method is globally convergent and locally Q-superlinearly convergent.*

Proof. [Proof to be filled in.] □

Remark 5.7 (BFGS in practice)**.** BFGS satisfies essentially all the items on the wish list: it is symmetric and positive definite (so $s^k = -B_k^{-1}\nabla f(x^k)$ is descent), close to B_k, cheap to update (in practice the Cholesky factors of B_k are stored and updated directly). Applied with exact linesearch to a strictly convex quadratic, B_k converges to $\nabla^2 f$ in at most n iterations. BFGS suffers much less from poor scaling than steepest descent. It has been the most popular general-purpose method when second derivatives are unavailable, and is the basis of MATLAB's `fminunc`.

5.3 Nonlinear least-squares

Many problems (model fitting, parameter estimation, data assimilation) take the form: choose x to minimise the sum of squared residuals $r_j(x)^2$. The problem is called *zero-residual* if $r(x^*) = 0$ (the model fits exactly) and *nonzero-residual* otherwise. The special structure of f allows significant savings over generic Newton methods.

The nonlinear least-squares problem is

$$\min_{x \in \mathbb{R}^n} f(x) = \tfrac{1}{2}\|r(x)\|^2 = \tfrac{1}{2}\sum_{j=1}^{m} r_j(x)^2, \tag{5.3}$$

where $r : \mathbb{R}^n \to \mathbb{R}^m$ is smooth with Jacobian $J(x)$.

Lemma 5.8 (Derivatives of f)**.** *For f in* (5.3)*,*

$$\nabla f(x) = J(x)^\top r(x), \qquad \nabla^2 f(x) = J(x)^\top J(x) + \sum_{j=1}^{m} r_j(x)\nabla^2 r_j(x).$$

Proof. [Proof to be filled in.] □

5.4 The Gauss-Newton method

Motivation. The Hessian decomposes as $\nabla^2 f(x) = J(x)^\top J(x) + \sum_j r_j(x)\nabla^2 r_j(x)$. When residuals are small ($r_j(x^*) \approx 0$) or the model is nearly linear ($\nabla^2 r_j \approx 0$), the second term is negligible and $\nabla^2 f(x) \approx J(x)^\top J(x)$. The Gauss-Newton method substitutes this approximation into the Newton system, dispensing with second derivatives entirely.

Definition 5.9 (Gauss-Newton direction)**.** The *Gauss-Newton (GN) direction* solves the linear system

$$J(x^k)^\top J(x^k)\, s^k = -J(x^k)^\top r(x^k). \tag{5.4}$$

Lemma 5.10 (Descent property)**.** *If $J(x^k)$ has full column rank, then the Gauss-Newton direction is a descent direction for f.*

Proof. [Proof to be filled in.] □

Theorem 5.11 (Convergence of Gauss-Newton)**.** *Apply Gauss-Newton with bArmijo linesearch.*

1. *(Global.) If $J(x^k)$ is uniformly full rank, $f \in \mathcal{C}^1$ bounded below and ∇f Lipschitz, then $\left\|\nabla f(x^k)\right\| \to 0$.*
2. *(Local.) If $r(x^*) = 0$ and $J(x^*)$ has full rank, the convergence is quadratic; if $r(x^*) \neq 0$ but small, the convergence is linear.*

Proof. [Proof to be filled in.] □

Remark 5.12 (Newton vs. Gauss-Newton)**.**

- **Cost per iteration:** Newton > Gauss-Newton (no second derivatives needed for GN).
- **Newton direction may be ascent** if $\nabla^2 f(x^k)$ is not positive definite. In contrast, the GN direction is always descent provided $J(x^k)$ has full column rank.
- **Local rate:** Newton always quadratic; GN only quadratic for zero-residual problems and only linear when $r(x^*) \neq 0$.
- Both methods are unreliable without a linesearch (or trust-region) safeguard.

Chapter 6

Trust-Region Methods

Trust-region methods are an alternative to linesearch for globalising Newton-type iterations. The two philosophies differ:

- **Linesearch:** "liberal" in choice of search direction; keeps bad behaviour in control by choosing α_k.
- **Trust-region:** "conservative" in choice of step; the full step is accepted if a sufficient *model decrease* is realised in the objective. Otherwise the step is rejected and the model is refined.

A key advantage of TR methods is that they accommodate *indefinite* or ill-behaved Hessians: the trust-region constraint regularises an otherwise unbounded model. They also fit naturally with quasi-Newton ideas, since there is no need for B_k to be positive definite.

6.1 The trust-region framework

We approximate $f(x^k + s)$ by a local model. Two natural choices are:

- linear: $\ell_k(s) := f(x^k) + s^\top \nabla f(x^k)$;
- quadratic: $q_k(s) := f(x^k) + s^\top \nabla f(x^k) + \frac{1}{2} s^\top B_k s$, with $B_k = \nabla^2 f(x^k)$ in the standard case.

Impediments. The model may not resemble f when $\|s\|$ is large, and may be unbounded below: ℓ_k is always unbounded below (unless $\nabla f(x^k) = 0$), and q_k is unbounded below whenever B_k is negative definite or indefinite, and sometimes when B_k is only positive semidefinite. We trust the model only on a region $\|s\| \le \Delta_k$: At iteration k, we form a quadratic model

$$m_k(s) = f(x^k) + s^\top \nabla f(x^k) + \tfrac{1}{2} s^\top B_k s, \tag{6.1}$$

where B_k is symmetric (e.g. $B_k = \nabla^2 f(x^k)$). The *trust-region (TR) subproblem* is

$$\min_{s \in \mathbb{R}^n} \; m_k(s) \quad \text{subject to} \quad \|s\| \le \Delta_k, \tag{TR}$$

with trust-region radius $\Delta_k > 0$.

Algorithm 6.1 (Generic Trust-Region (GTR) method)**.** Given $\Delta_0 > 0$, $x^0 \in \mathbb{R}^n$, $\varepsilon > 0$. While $\left\|\nabla f(x^k)\right\| > \varepsilon$:

1. Form m_k and (approximately) solve (TR) for s^k with $m_k(s^k) < f(x^k)$.
2. Compute $\rho_k = \dfrac{f(x^k) - f(x^k + s^k)}{f(x^k) - m_k(s^k)}$.

3. If $\rho_k \geq 0.9$: set $x^{k+1} := x^k + s^k$, $\Delta_{k+1} := 2\Delta_k$.
   Else if $\rho_k \geq 0.1$: set $x^{k+1} := x^k + s^k$, $\Delta_{k+1} := \Delta_k$.
   Else: set $x^{k+1} := x^k$, $\Delta_{k+1} := \frac{1}{2}\Delta_k$.

Remark 6.2 (The ratio ρ_k)**.** ρ_k measures the ratio of the *actual function decrease* to the *predicted model decrease.* $\rho_k \approx 1$ indicates the model is accurate (we expand the trust region to take longer steps); $\rho_k \ll 1$ indicates the model is poor (we reject the step and shrink the trust region). The thresholds $0.1, 0.9$ are conventional; other reasonable values appear in practice.

6.2 The Cauchy point

The Cauchy point is the trust-region analogue of the steepest descent direction: it minimises the model m_k along the steepest descent direction subject to the trust-region constraint. Recall (Theorem 3.3) that steepest descent has strong global convergence properties; insisting that any approximate solution s^k of the TR subproblem do at least as well as the Cauchy point is a "minimal sufficient decrease" condition that suffices for global convergence of GTR.

Definition 6.3 (Cauchy point)**.** The *Cauchy point* of (TR) is $y_c^k = x^k + s_c^k$, where $s_c^k = -\alpha_c^k \nabla f(x^k)$ and

$$\alpha_c^k = \arg\min_{\alpha > 0} m_k\Big(-\alpha \nabla f(x^k)\Big) \quad \text{subject to} \quad \Big\|\alpha \nabla f(x^k)\Big\| \leq \Delta_k.$$

Lemma 6.4 (Cauchy-point decrease)**.** *With* $h_k = \nabla f(x^k)^\top B_k \nabla f(x^k)$,

$$f(x^k) - m_k(s_c^k) \geq \tfrac{1}{2}\Big\|\nabla f(x^k)\Big\| \min\left\{\Delta_k, \frac{\Big\|\nabla f(x^k)\Big\|}{\|B_k\|}\right\}.$$

Proof. [Proof to be filled in.] □

Remark 6.5. Lemma 6.4 is the key technical estimate underlying global convergence: any step that does at least as well as the Cauchy step inherits this lower bound on the model decrease, and combined with the ratio test in GTR this drives $\Big\|\nabla f(x^k)\Big\| \to 0$.

6.3 Global convergence of GTR

Theorem 6.6 (Global convergence of GTR)**.** *Let* $f \in \mathcal{C}^2(\mathbb{R}^n)$ *be bounded below with* ∇f *Lipschitz continuous, and assume* $\|B_k\|$ *is uniformly bounded. Apply GTR (Algorithm 6.1) requiring* $m_k(s^k) \leq m_k(s_c^k)$*. Then either there exists* $\ell \geq 0$ *with* $\nabla f(x^\ell) = 0$*, or* $\liminf_{k\to\infty} \Big\|\nabla f(x^k)\Big\| = 0$.

Proof. [Proof to be filled in.] □

Theorem 6.7 (Stronger global convergence)**.** *Under the conditions of Theorem 6.6,* $\lim_{k\to\infty} \Big\|\nabla f(x^k)\Big\| = 0$.

Proof. [Proof to be filled in.] □

6.4 Solving the trust-region subproblem

Strategy. The characterisation below shows that solving (TR) reduces to the secular equation $\Big\|(B_k + \lambda I)^{-1}\nabla f(x^k)\Big\| = \Delta_k$ in the scalar $\lambda \geq \max\{0, -\lambda_{\min}(B_k)\}$.

Theorem 6.8 (Characterization of the global TR solution)**.** *A vector $s^* \in \mathbb{R}^n$ is a global solution of* (TR) *if and only if $\|s^*\| \le \Delta_k$ and there exists $\lambda^* \ge 0$ such that*

1. $(B_k + \lambda^* I)s^* = -\nabla f(x^k)$,
2. $\lambda^*(\Delta_k - \|s^*\|) = 0$ *(complementarity),*
3. $B_k + \lambda^* I \succeq 0$.

Moreover λ^* is unique whenever $B_k + \lambda^* I \succ 0$.

Proof. [Proof to be filled in.] □

Remark 6.9 (Two cases for the global solution)**.** Theorem 6.8 reduces practical solution to two cases:

- **Case 1:** $B_k \succ 0$ and the unconstrained Newton step $s = -B_k^{-1}\nabla f(x^k)$ satisfies $\|s\| \le \Delta_k$ — then $s^* = s$ and $\lambda^* = 0$ (Newton step is inside the trust region).
- **Case 2:** otherwise — find $\lambda \ge \max\{0, -\lambda_{\min}(B_k)\}$ with $\left\|(B_k + \lambda I)^{-1}\nabla f(x^k)\right\| = \Delta_k$, by a 1-D root-find; the solution s^* lies on the boundary.

Remark 6.10 (Large-scale problems)**.** For large n, repeated Cholesky factorisations of $B_k + \lambda I$ become prohibitive. In practice one uses iterative methods (truncated conjugate gradient, Lanczos / Steihaug–Toint) that produce an approximate solution at least as good as the Cauchy point, hence preserving the global convergence guarantee.

Remark 6.11 (Linesearch vs. trust-region in practice)**.** Both linesearch and trust-region methods underpin state-of-the-art unconstrained optimisation software. Linesearch needs more heuristics to deal with negative curvature; TR handles indefinite Hessians naturally and accommodates quasi-Newton B_k that need not be positive definite. The choice between them is largely a matter of taste.

Chapter 7

Optimality Conditions for Constrained Optimization

The unconstrained theory characterised optimality by $\nabla f(x^*) = 0$. For constrained problems this fails: the gradient at the minimiser need not vanish, because feasibility is preventing further descent. The right generalisation is the *Karush–Kuhn–Tucker (KKT)* conditions, which characterise points at which no *feasible* descent direction exists.

The development proceeds in three steps:

1. Define geometric (tangent cone T_Ω) and algebraic (linearised feasible directions $\mathcal{F}$) descriptions of feasible directions; always $T_\Omega \subseteq \mathcal{F}$, but the reverse can fail.
2. Identify a *constraint qualification (CQ)* ensuring $T_\Omega = \mathcal{F}$ — without a CQ, optimality conditions need not hold.
3. Derive the KKT conditions from the algebraic description $\mathcal{F}$.

7.1 The constrained problem

We consider the general constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c_i(x) = 0,\ i \in E;\ \ c_i(x) \geq 0,\ i \in I, \tag{CP}$$

with feasible set $\Omega = \{x \in \mathbb{R}^n : c_i(x) = 0,\ i \in E;\ c_i(x) \geq 0,\ i \in I\}$, and active set at $x \in \Omega$: $\mathcal{A}(x) = E \cup \{i \in I : c_i(x) = 0\}$.

7.2 Tangent cone and linearized feasible directions

Definition 7.1 (Tangent cone)**.** The *tangent cone* to Ω at $x \in \Omega$ is

$$T_\Omega(x) = \Big\{ s \in \mathbb{R}^n : \exists\, z^k \in \Omega,\ t^k \downarrow 0,\ z^k \to x,\ s = \lim_{k\to\infty} \frac{z^k - x}{t^k} \Big\}.$$

Definition 7.2 (Linearized feasible directions)**.** The set of *linearized feasible directions* at $x \in \Omega$ is

$$\mathcal{F}(x) = \{ s : \nabla c_i(x)^\top s = 0,\ i \in E;\ \nabla c_i(x)^\top s \geq 0,\ i \in I \cap \mathcal{A}(x) \}.$$

Lemma 7.3. $T_\Omega(x) \subseteq \mathcal{F}(x)$.

Proof. [Proof to be filled in.] □

7.3 Constraint qualifications

Definition 7.4 (LICQ)**.** (CP) satisfies the *Linear Independence Constraint Qualification (LICQ)* at $x \in \Omega$ if the active constraint gradients $\{\nabla c_i(x) : i \in \mathcal{A}(x)\}$ are linearly independent.

Definition 7.5 (Slater CQ)**.** A convex problem (CP) (with $c_E(x) = Ax - b$, c_i concave for $i \in I$) satisfies the *Slater CQ* if there exists $\bar{x}$ with $A\bar{x} = b$ and $c_i(\bar{x}) > 0$ for all $i \in I$.

Theorem 7.6 (CQ implies $T_\Omega = \mathcal{F}$)**.** *Under LICQ (or any standard CQ), $T_\Omega(x) = \mathcal{F}(x)$.*

Proof. [Proof to be filled in.] □

Remark 7.7 (Linear constraints)**.** If the constraints of (CP) are linear, i.e. $(c_E, c_I)(x) = Ax - b$, then *no constraint qualification is required*: the KKT conditions hold at every local minimiser. This is the content of Theorem 17 in the slides.

7.4 First-order necessary conditions (KKT)

Interpretation. The KKT conditions generalise the unconstrained stationarity condition $\nabla f = 0$: at a constrained minimiser, ∇f lies in the cone generated by the gradients of active constraints (with appropriate signs). Inactive inequality constraints contribute nothing — by complementarity $\lambda_i^* c_i(x^*) = 0$, so $c_i(x^*) > 0$ forces $\lambda_i^* = 0$.

Definition 7.8 (Lagrangian)**.** The *Lagrangian* of (CP) is

$$\mathcal{L}(x, y, \lambda) = f(x) - \sum_{i \in E} y_i c_i(x) - \sum_{i \in I} \lambda_i c_i(x).$$

Theorem 7.9 (KKT first-order necessary conditions)**.** *Let x^* be a local minimizer of* (CP) *at which a CQ holds. Then there exist multipliers $y^* \in \mathbb{R}^{|E|}$ and $\lambda^* \in \mathbb{R}^{|I|}$ such that*

$$\begin{aligned}
\nabla f(x^*) &= \sum_{i \in E} y_i^* \nabla c_i(x^*) + \sum_{i \in I} \lambda_i^* \nabla c_i(x^*), \\
c_i(x^*) &= 0, \quad i \in E, \\
c_i(x^*) &\geq 0, \quad i \in I, \\
\lambda_i^* &\geq 0, \quad i \in I, \\
\lambda_i^* c_i(x^*) &= 0, \quad i \in I \quad \textit{(complementarity)}.
\end{aligned}$$

Proof. [Proof to be filled in.] □

7.5 Second-order conditions

Let $\mathcal{F}(\lambda^*) = \{s \in \mathcal{F}(x^*) : \nabla c_i(x^*)^\top s = 0 \text{ for } i \in I \text{ with } \lambda_i^* > 0\}$.

Theorem 7.10 (Second-order necessary conditions)**.** *Let x^* be a local minimizer of* (CP) *at which a CQ holds, with multipliers (y^*, λ^*). Then $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) s \geq 0$ for all $s \in \mathcal{F}(\lambda^*)$.*

Proof. [Proof to be filled in.] □

Theorem 7.11 (Second-order sufficient conditions)**.** *Let x^* be feasible for* (CP)*, and let (y^*, λ^*) satisfy the KKT conditions at x^*. If $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) s > 0$ for all $0 \neq s \in \mathcal{F}(\lambda^*)$, then x^* is a strict local minimizer of* (CP)*.*

Proof. [Proof to be filled in.] □

7.6 Convex problems

Theorem 7.12 (KKT is sufficient for convex problems)**.** *Suppose* (CP) *is a convex problem (i.e. f convex, c_i concave for $i \in I$, c_E affine). If $\hat{x}$ is a KKT point of* (CP)*, then $\hat{x}$ is a global minimizer of* (CP)*.*

Proof. [Proof to be filled in.] □

Chapter 8

Penalty Methods

Constrained optimisation introduces a fundamental tension: we want to minimise f *and* satisfy the constraints. For inequality-constrained or linear equality-constrained problems we can sometimes maintain feasibility throughout, but with general nonlinear equality constraints this is very hard, even impossible. The idea behind *penalty methods* is to replace the constrained problem by a sequence of unconstrained problems whose minimisers approach the original solution as a parameter is varied.

We now turn to algorithms for the equality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) = 0, \tag{eCP}$$

where $c = (c_1, \ldots, c_m) : \mathbb{R}^n \to \mathbb{R}^m$ is smooth.

8.1 The quadratic penalty function

Definition 8.1 (Quadratic penalty function)**.** For $\sigma > 0$, the *quadratic penalty function* associated with (eCP) is

$$\Phi_\sigma(x) = f(x) + \frac{1}{2\sigma} \|c(x)\|^2 . \tag{8.1}$$

Interpretation. The parameter $\sigma > 0$ controls the penalty on infeasibility. As $\sigma \to 0$, infeasibility is forced down and the unconstrained minimisers of Φ_σ approach a solution of (eCP). The function Φ_σ may have stationary points that are *not* solutions of (eCP) — for example, local minimisers of $\|c(x)\|$ when $c(x) = 0$ has no solution.

Algorithm 8.2 (Basic quadratic penalty method)**.** Choose $\sigma_0 > 0$ and $\varepsilon_0 > 0$. For $k = 0, 1, 2, \ldots$:

1. Find x^k with $\left\|\nabla \Phi_{\sigma_k}(x^k)\right\| \le \varepsilon_k$.
2. Choose $\sigma_{k+1} < \sigma_k$ and $\varepsilon_{k+1} < \varepsilon_k$.

Lemma 8.3 (Gradient of Φ_σ)**.** *With $J(x)$ the Jacobian of c,*

$$\nabla \Phi_\sigma(x) = \nabla f(x) + \frac{1}{\sigma} J(x)^\top c(x) = \nabla f(x) - J(x)^\top y(x),$$

where $y(x) = -c(x)/\sigma$.

Proof. [Proof to be filled in.] □

8.2 Convergence of the penalty method

Theorem 8.4 (Global convergence of the quadratic penalty method)**.** *Apply Algorithm 8.2 to* (eCP)*. Assume* $f, c \in \mathcal{C}^1$*, set* $y_i^k = -c_i(x^k)/\sigma_k$*, and assume* $\left\|\nabla\Phi_{\sigma_k}(x^k)\right\| \le \varepsilon_k$ *with* $\varepsilon_k \to 0$ *and* $\sigma_k \to 0$ *as* $k \to \infty$*. Suppose further that* $x^k \to x^*$ *where* $\{\nabla c_i(x^*) : i = 1, \dots, m\}$ *are linearly independent. Then* x^* *is a KKT point of* (eCP) *and* $y^k \to y^*$*, the corresponding Lagrange multiplier vector.*

Proof. [Proof to be filled in.] □

Remark 8.5. If $J(x^*)$ is not full rank then x^* instead (locally) minimizes the infeasibility $\|c(x)\|$.

Remark 8.6 (Ill-conditioning of $\nabla^2\Phi_{\sigma_k}$)**.** A serious practical drawback of the basic penalty method is that, asymptotically, $\nabla^2\Phi_{\sigma_k}(x^k)$ has m eigenvalues of order $O(1/\sigma_k)$ and the remaining $n-m$ eigenvalues of order $O(1)$ — so the condition number is $O(1/\sigma_k)$ and *blows up as* $\sigma_k \to 0$. Consequently we may not be able to compute x^k accurately by Newton's method on Φ_{σ_k}. This motivates augmented Lagrangians (Chapter 9), which avoid the need to drive $\sigma_k \to 0$.

8.3 Other penalty functions

The quadratic penalty is *smooth* but *inexact*: $\sigma_k \to 0$ is required, with the ill-conditioning consequences in Remark 8.6. *Exact* penalty functions allow convergence at a fixed (sufficiently small) σ, avoiding ill-conditioning, but are typically non-smooth and so harder to minimise.

Definition 8.7 (Exact penalty function)**.** A penalty function $\Phi(x, \sigma)$ is *exact* if there exists $\sigma^* > 0$ such that for all $\sigma < \sigma^*$, every local solution of the original problem is a local minimizer of $\Phi(\cdot, \sigma)$.

Example 8.8 (ℓ_1 and ℓ_2 penalty functions)**.** For (CP),

$$\Phi_1(x, \sigma) = f(x) + \tfrac{1}{\sigma}\sum_{i\in E}|c_i(x)| + \tfrac{1}{\sigma}\sum_{i\in I}[c_i(x)]_-,$$

$$\Phi_2(x, \sigma) = f(x) + \tfrac{1}{\sigma}\|c_E(x)\|_2 + \tfrac{1}{\sigma}\|[c_I(x)]_-\|_2 .$$

Both are exact (but generally non-smooth); the quadratic penalty is smooth but not exact.

Chapter 9

Augmented Lagrangian Methods

The quadratic penalty method has a fundamental drawback: $\sigma_k \to 0$ is required, leading to severe ill-conditioning of the subproblem Hessian (Remark 8.6). The augmented Lagrangian method retains the quadratic penalty term but adds an explicit estimate $u \in \mathbb{R}^m$ of the Lagrange multipliers, so that as $u \to y^*$ feasibility is achieved *without* driving σ to zero.

9.1 The augmented Lagrangian

Definition 9.1 (Augmented Lagrangian)**.** The *augmented Lagrangian* for (eCP) is

$$\Phi(x, u, \sigma) = f(x) - u^\top c(x) + \frac{1}{2\sigma} \|c(x)\|^2 , \tag{9.1}$$

where $u \in \mathbb{R}^m$ is an estimate of the Lagrange multipliers and $\sigma > 0$.

Remark 9.2 (Two interpretations of $\Phi(x, u, \sigma)$)**.** $\Phi(x, u, \sigma)$ admits two complementary readings:

- **Lagrangian augmented with a quadratic penalty**, $\mathcal{L}(x, u) + \frac{1}{2\sigma} \|c(x)\|^2$. The quadratic term *convexifies* the Lagrangian locally, fixing the well-known issue that the Lagrangian itself is generally a saddle in (x, y) rather than a minimum in x.
- **Shifted quadratic penalty**: completing the square in $c(x)$ gives $\Phi(x, u, \sigma) = f(x) + \frac{1}{2\sigma} \|c(x) - \sigma u\|^2 - \frac{\sigma}{2} \|u\|^2$, a quadratic penalty centred at the shifted target $c(x) = \sigma u$ rather than $c(x) = 0$. As $u \to y^*$, the shift vanishes correctly even with σ bounded away from zero.

The aim is to adjust both u and σ to encourage convergence.

Lagrange multiplier estimates. A short calculation gives

$$\nabla_x \Phi(x, u, \sigma) = \nabla f(x) - J(x)^\top y(x), \qquad y(x) := u - c(x)/\sigma,$$

so the gradient of the augmented Lagrangian coincides with the gradient of the ordinary Lagrangian evaluated at the multiplier estimate $y(x)$. This is the update rule used in Algorithm 9.3.

9.2 The basic algorithm

Algorithm 9.3 (Augmented Lagrangian method)**.** Choose $u^0 \in \mathbb{R}^m$, $\sigma_0 > 0$, $\varepsilon_0 > 0$. For $k = 0, 1, \ldots$:

1. Find x^k with $\left\|\nabla_x \Phi(x^k, u^k, \sigma_k)\right\| \le \varepsilon_k$.
2. Update $u^{k+1} = u^k - c(x^k)/\sigma_k$.
3. Choose $\sigma_{k+1} \le \sigma_k$, $\varepsilon_{k+1} < \varepsilon_k$.

9.3 Convergence

Theorem 9.4 (Global convergence of augmented Lagrangian)**.** *Apply Algorithm 9.3 to* (eCP)*. Assume* $f, c \in \mathcal{C}^1$*, let* $y^k = u^k - c(x^k)/\sigma_k$*, and* $\left\|\nabla_x \Phi(x^k, u^k, \sigma_k)\right\| \leq \varepsilon_k$ *with* $\varepsilon_k \to 0$*. Suppose* $x^k \to x^*$ *where* $\{\nabla c_i(x^*)\}_{i=1}^m$ *are linearly independent. Then* $y^k \to y^*$ *with* $\nabla f(x^*) - J(x^*)^\top y^* = 0$*. Moreover, if either* $\sigma_k \to 0$ *for bounded* u^k*, or* $u^k \to y^*$ *for bounded* σ_k*, then* x^* *is a KKT point of* (eCP) *with associated multipliers* y^**.*

Proof. [Proof to be filled in.] □

Remark 9.5 (Why the augmented Lagrangian helps)**.** The crucial difference from the basic penalty method is the second alternative in Theorem 9.4: convergence is achieved as soon as $u^k \to y^*$*, with* σ_k *bounded away from zero.* This means the ill-conditioning of the subproblem Hessian is controlled — $\nabla^2_{xx}\Phi(x, u, \sigma) = \nabla^2_{xx}\mathcal{L}(x, y) + \frac{1}{\sigma} J(x)^\top J(x)$ has condition number $O(1/\sigma)$, which stays bounded if σ does. Augmented Lagrangian methods underpin the LANCELOT software and are competitive with state-of-the-art nonlinear programming codes.

Chapter 10

Interior Point Methods

For inequality-constrained problems, an alternative to penalty methods is to *stay strictly inside the feasible region throughout* and rely on a *barrier* that diverges as $c_i(x) \downarrow 0$. As we relax the barrier parameter μ, the unconstrained minimisers approach the boundary along the so-called *central path*.

We consider the inequality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) \geq 0, \tag{iCP}$$

where $c = (c_1, \ldots, c_p) : \mathbb{R}^n \to \mathbb{R}^p$ is smooth.

10.1 The logarithmic barrier function

Why the logarithm? Since $-\log c_i(x) \to +\infty$ as $c_i(x) \downarrow 0$, any minimiser $x(\mu)$ of f_μ must lie strictly inside Ω — the constraint $c(x) > 0$ is enforced *implicitly* rather than as an explicit inequality. When μ is large, $x(\mu)$ sits "far inside" Ω; when μ is small, f dominates and $x(\mu)$ is close to the optimal boundary of Ω (this is also the regime in which ill-conditioning kicks in; see below).

Definition 10.1 (Barrier function)**.** For $\mu > 0$, the *logarithmic barrier function* is

$$f_\mu(x) = f(x) - \mu \sum_{i=1}^{p} \log c_i(x), \tag{10.1}$$

defined on the strict interior $\{x : c(x) > 0\}$.

Lemma 10.2 (Barrier derivatives)**.** *With $J(x)$ the Jacobian of c and $C(x) = \operatorname{diag}(c_1(x), \ldots, c_p(x))$,*

$$\nabla f_\mu(x) = \nabla f(x) - \mu J(x)^\top c^{-1}(x) = \nabla f(x) - J(x)^\top \lambda(x),$$
$$\nabla^2 f_\mu(x) = \nabla^2 f(x) - \sum_{i=1}^{p} \frac{\mu}{c_i(x)} \nabla^2 c_i(x) + \mu J(x)^\top C(x)^{-2} J(x),$$

where $\lambda_i(x) := \mu / c_i(x)$ are estimates of the Lagrange multipliers.

Proof. [Proof to be filled in.] □

Remark 10.3 (Connection to the Lagrangian)**.** Lemma 10.2 reveals a striking fact: $\nabla f_\mu(x) = \nabla_x \mathcal{L}(x, \lambda(x))$, the gradient of the Lagrangian of (iCP) evaluated at the multiplier estimate $\lambda(x) = \mu / c(x)$. Hence f_μ stationarity coincides with stationarity of the Lagrangian of the original problem, with multipliers $\lambda(x)$. As $\mu \to 0$ the conditions $\lambda_i(x) c_i(x) = \mu \to 0$ recover complementarity.

10.2 The central path

Theorem 10.4 (Local existence of the central path)**.** *Assume $\Omega^o \neq \emptyset$ and x^* is a local minimiser of* (iCP) *satisfying:*

1. $\lambda_i^* > 0$ *whenever* $c_i(x^*) = 0$ (strict complementarity)*;*
2. $\{\nabla c_i(x^*) : i \in \mathcal{A}\}$ *are linearly independent (LICQ);*
3. *there exists* $\alpha > 0$ *such that* $s^\top \nabla^2_{xx}\mathcal{L}(x^*, \lambda^*)s \geq \alpha \|s\|^2$ *for all* s *with* $J_{\mathcal{A}}(x^*)s = 0$.

Then there exists a unique, continuously differentiable path $\{x(\mu) : \mu_\varepsilon > \mu > 0\}$ *of (global) minimisers of* f_μ*, with* $x(\mu) \to x^*$ *as* $\mu \to 0$.

Proof. [Proof to be filled in.] □

10.3 The basic barrier algorithm

Algorithm 10.5 (Basic barrier algorithm)**.** Choose $\mu_0 > 0$, $\varepsilon_0 > 0$ and x^0 with $c(x^0) > 0$. For $k = 0, 1, \ldots$:

1. Find x^k with $c(x^k) > 0$ and $\left\|\nabla f_{\mu_k}(x^k)\right\| \leq \varepsilon_k$.
2. Choose $\mu_{k+1} < \mu_k$, $\varepsilon_{k+1} < \varepsilon_k$.

10.4 Convergence

Theorem 10.6 (Global convergence of the barrier method)**.** *Apply Algorithm 10.5 to* (iCP)*. Assume* $f, c \in \mathcal{C}^1$*,* $\lambda_i^k = \mu_k / c_i(x^k)$ *for* $i = 1, \ldots, p$ *with* $c(x^k) > 0$*,* $\mu_k > 0$*, and* $\left\|\nabla f_{\mu_k}(x^k)\right\| \leq \varepsilon_k$ *with* $\varepsilon_k \to 0$*,* $\mu_k \to 0$ *as* $k \to \infty$*. Suppose* $x^k \to x^*$ *where the active gradients* $\{\nabla c_i(x^*) : i \in \mathcal{A}\}$ *are linearly independent (i.e. LICQ),* $\mathcal{A} = \{i : c_i(x^*) = 0\}$*. Then* x^* *is a KKT point of* (iCP) *and* $\lambda^k \to \lambda^*$*, the associated Lagrange multipliers.*

Proof. [Proof to be filled in.] □

Remark 10.7 (Difficulties of basic (primal) IPMs)**.**

- *Ill-conditioning of* $\nabla^2 f_\mu$*.* As $\mu \to 0$, $\mu J(x)^\top C(x)^{-2} J(x)$ has $|\mathcal{A}|$ eigenvalues of order $O(1/\mu)$ tending to infinity, while the rest are $O(1)$. The condition number is $O(1/\mu_k)$, blowing up. This is the main reason barrier methods fell out of favour in the 1960s, until primal-dual methods revived them in the 1980s.
- *Poor starting points.* The current iterate x^k is generally a poor starting point for the next subproblem $\min f_{\mu_{k+1}}$: a full Newton step typically violates $c(x^k + s^k) > 0$ as soon as μ is meaningfully decreased, so progress is slow.

The remedy for both is to switch to primal-dual methods.

10.5 Primal-dual interior point methods

Remark 10.8 (Primal-dual reformulation)**.** Modern interior point software (KNITRO, IPOPT, GALAHAD) works with the *perturbed KKT system*

$$\nabla f(x) - J(x)^\top \lambda = 0, \qquad C(x)\lambda = \mu e, \qquad c(x) > 0,\ \lambda > 0,$$

where $e = (1, \ldots, 1)^\top$. The path traced by solutions as $\mu \downarrow 0$ is the *central path.* Newton's method applied to the perturbed KKT system treats (x, λ) as joint variables — multiplier

updates are computed explicitly each iteration, in contrast to the implicit estimate $\lambda(x) = \mu/c(x)$ in the primal method.

Benefits over the primal barrier method:

- Ill-conditioning is sidestepped: solving the linear system in the right (x, λ) subspaces, the asymptotic conditioning is benign.
- Current iterates are good starting points for the next subproblem, so only a few Newton iterations per μ_k are needed.
- Aggressive parameter schedule $\mu_{k+1} = O(\mu_k^2)$ yields *superlinear* asymptotic convergence.

Computing an initial x^0 with $c(x^0) > 0$ is itself nontrivial; various heuristics exist (e.g. phase-I problems).

Remark 10.9 (Linear programming: simplex vs. IPMs)**.** For LP, the simplex method has worst-case exponential complexity (Klee–Minty 1972) but very good average-case behaviour. Karmarkar's IPM (1984) gave the first practical polynomial-time algorithm for LP; the central path is unique and global, and Newton's method on the barrier function can be precisely quantified. IPMs scale to very large LPs, with numerically observed average complexity $O(\log(\text{LP dim}))$ iterations, each more expensive than a simplex pivot.

Appendix A

Useful background

A.1 Taylor's theorem

Theorem A.1 (First-order Taylor)**.** *Let $f \in \mathcal{C}^1$. For any $x, s \in \mathbb{R}^n$ there exists $\tilde{\alpha} \in (0,1)$ with $f(x+s) = f(x) + \nabla f(x + \tilde{\alpha} s)^\top s$.*

Theorem A.2 (Second-order Taylor)**.** *Let $f \in \mathcal{C}^2$. For any $x, s \in \mathbb{R}^n$ there exists $\tilde{\alpha} \in (0,1)$ with*

$$f(x+s) = f(x) + \nabla f(x)^\top s + \tfrac{1}{2} s^\top \nabla^2 f(x + \tilde{\alpha} s) s.$$

A.2 Lipschitz continuity

Definition A.3. A function $g : \mathbb{R}^n \to \mathbb{R}^m$ is *Lipschitz continuous* on $\mathbb{R}^n$ with constant $L > 0$ if $\|g(y) - g(x)\| \leq L \|y - x\|$ for all $x, y \in \mathbb{R}^n$.

A.3 Convexity

Theorem A.4 (First-order characterization)**.** *Let $f \in \mathcal{C}^1$. Then f is convex if and only if $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$ for all x, y.*

Theorem A.5 (Second-order characterization)**.** *Let $f \in \mathcal{C}^2$. Then f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all x.*